Adapt in the Wild: Test-Time Entropy Minimization with Sharpness and Feature Regularization

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***Abstract*—Test-time adaptation (TTA) has shown to be effective at tackling distribution shifts between training and testing data by adapting a given model on test samples. However, the online model updating of TTA may be unstable and this is often a key obstacle preventing existing TTA methods from being deployed in the real world. Specifically, TTA may fail to improve or even harm the model performance when test data have: 1) mixed distribution shifts, 2) small batch sizes, and 3) online imbalanced label distribution shifts, which are quite common in practice. In this paper, we investigate the unstable reasons and find that the batch norm layer is a crucial factor hindering TTA stability. Conversely, TTA can perform more stably with batch-agnostic norm layers, *i.e.*, group or layer norm. However, we observe that TTA with group and layer norms does not always succeed and still suffers many failure cases, *i.e.*, the model collapses into trivial solutions by assigning the same class label for all samples. By digging into this, we find that, during the collapse process: 1) the model gradients often undergo an initial explosion followed by rapid degradation, suggesting that certain noisy test samples with large gradients may disrupt adaptation; and 2) the model representations tend to exhibit high correlations and classification bias. To address the above collapse issue, we first propose a sharpness-aware and reliable entropy minimization method, called SAR, for stabilizing TTA from two aspects: 1) remove partial noisy samples with large gradients, 2) encourage model weights to go to a flat minimum so that the model is robust to the remaining noisy samples. Based on SAR, we further introduce SAR^2 to prevent representation collapse with two regularizers: 1) a redundancy regularizer, which reduces inter-dimensional correlations among centroid-invariant features; and 2) an inequity regularizer, which maximizes the prediction entropy of a prototype centroid, thereby penalizing biased representations toward any specific class. Promising results demonstrate that our methods perform more stably over prior methods and are computationally efficient under the above wild test scenarios.**

***Index Terms*—Out-of-distribution generalization, test-time adaptation, sharpness-aware minimization, feature regularization.**

I. Introduction

DEEP neural networks achieve excellent performance when training and testing domains follow the same distribution [1], [2], [3]. However, when domain shifts exist, deep networks often struggle to generalize. Such domain shifts usually occur in real applications, since test data may unavoidably encounter natural variations or corruptions [4], [5], such as the weather changes (*e.g.*, *snow, frost, fog*), sensor degradation (*e.g.*, *Gaussian noise, defocus blur*), and many other reasons. Unfortunately, deep models can be sensitive to the above shifts and suffer from severe performance degradation even if the shift is mild [6]. However, deploying a deep model on test domains with distribution shifts is still an urgent demand, and model adaptation is needed in these cases.

Recently, numerous test-time adaptation (TTA) methods [7], [8], [9], [10] have been proposed to conquer the above domain shifts by online updating a model on the test data, which include two main categories, *i.e.*, Test-Time Training (TTT) [7], [11] and Fully TTA [8], [12]. In this work, we focus on Fully TTA since it is more generally to be used than TTT in two aspects: i) it does not alter training and can adapt arbitrary pre-trained models to the test data without access to original training data; ii) it may rely on fewer backward passes (only one or less than one) for each test sample than TTT (see efficiency comparisons of TTT, Tent and EATA in Table II).

TTA has been shown boost model robustness to domain shifts significantly. However, its excellent performance is often obtained under some mild test settings, *e.g.*, adapting with a batch of test samples that have the same distribution shift type and randomly shuffled label distribution (see Fig. 1 ①). In the complex real world, test data may come arbitrarily. As shown

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The source code is available at https://github.com/mr-eggplant/SAR.

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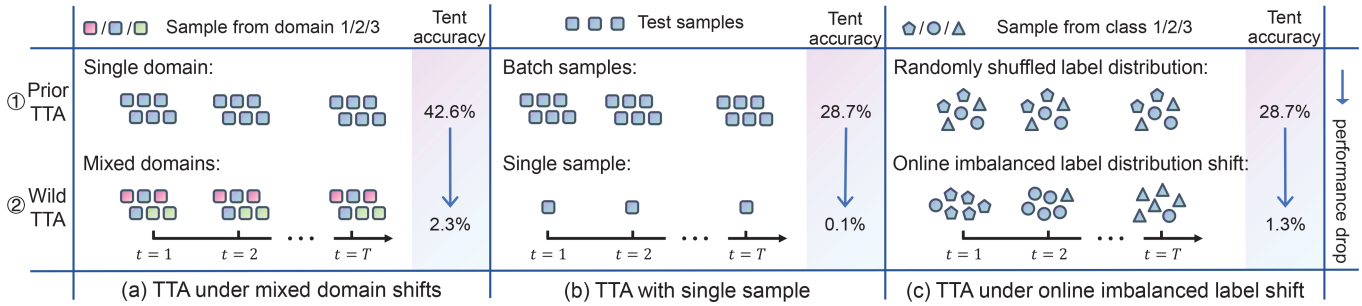


Fig. 1. An illustration of practical/wild test-time adaptation (TTA) scenarios, in which prior online TTA methods may degrade severely. The accuracy of Tent [8] is measured on ImageNet-C of level 5 with ResNet50-BN (15 mixed corruptions in *(a)* and Gaussian in *(b-c)*).

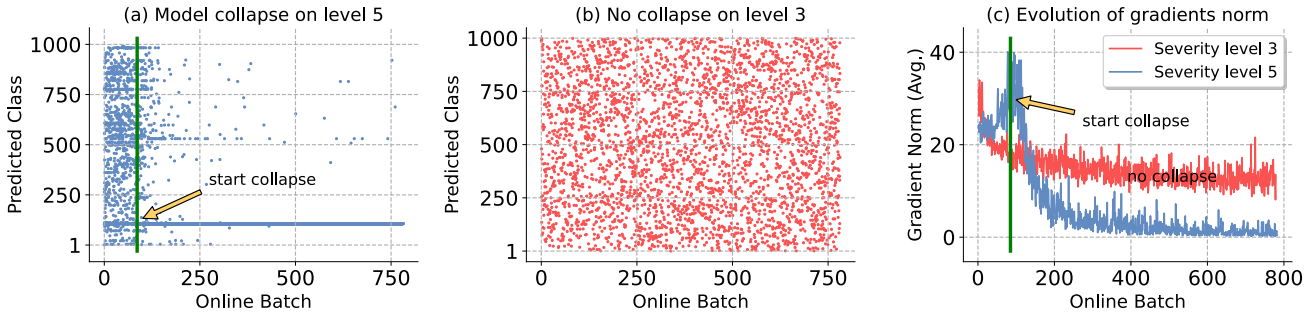


Fig. 2. Failure case analyses of online test-time entropy minimization [8]. (a) and (b) record the model predictions during online adaptation. At severity level 3, the model adapts normally, and its predictions are distributed across the label space. In contrast, at severity level 5, the model suffers from collapse and predicts almost all inputs as a specific class, regardless of their true labels. (c) illustrates how the gradient norm evolves with and without model collapse. When collapse occurs, the gradient norm first increases sharply and then quickly decreases to nearly zero. All experiments are conducted on shuffled ImageNet-C of Gaussian noise with ResNet50 (GN), and a larger (severity) level denotes a more severe distribution shift.

in Fig. 1 ②, the test scenario may meet: i) mixture of multiple distribution shifts, ii) small test batch sizes (even single sample), iii) the ground-truth test label distribution $Q_t(y)$ is online shifted and $Q_t(y)$ may be imbalanced at each time-step t. In these wild test settings, online updating a model by existing TTA methods may be unstable, *i.e.*, failing to help or even harming the model's robustness.

In this paper, we first point out that the batch norm (BN) layer [13] is a key obstacle since the mean and variance estimation in BN layers will be biased under the above wild scenarios. In light of this, we investigate the effects of norm layers in TTA (see Section VI) and find that pre-trained models with batch-agnostic norm layers (*i.e.*, group norm (GN) [14] and layer norm (LN) [15]) are more beneficial for stable TTA. However, TTA on GN/LN models does not always succeed and still has many failure cases. Specifically, GN/LN models optimized by online entropy minimization [8] tend to occur collapse, *i.e.*, predicting all samples to a single class (see Fig. 2), especially when the distribution shift is severe. To address this issue, we propose a **s**harpness-**a**ware and **r**eliable entropy minimization method (namely SAR). Specifically, we find that indeed some noisy samples that produce gradients with large norms harm the adaptation and thus result in model collapse. To avoid this, we filter test samples with large and noisy gradients out of adaptation according to their entropy. For the remaining samples, we introduce a sharpness-aware learning scheme to ensure that the model weights are optimized to a flat minimum, thereby being robust to the large and noisy gradients/updates.

Recently, our observations have been widely adopted in the TTA community [16], [17], [18] and SAR has inspired a series of follow-up studies for TTA in the wild, including DeYO [19] that selects samples based on shape information for model updates, ROID [20] that directly adjusts prediction priors under imbalanced shift, *etc.* Nevertheless, they still analyze and devise methods to conquer the collapse issue at the model's output level, while the intrinsic factors driving model collapse remain underexplored.

Therefore, we further move the lens one level deeper and ask: *What happens inside the model while collapse is (or is not) unfolding?* We start by analyzing the representation dynamics of model throughout TTA. Specifically, we define two feature-level statistics: 1) feature redundancy to estimate dimension correlation; and 2) feature inequity that calculates the prediction entropy regarding a central feature, which quantifies the bias in representation. We compare the trajectories of these two statistics during a collapse and a non-collapse TTA process in Fig. 3. Our results uncover a strong correlation between feature redundancy/inequity and TTA effectiveness. Feature redundancy declines or remains stable when TTA effectively enhances accuracy, but explodes immediately when collapse occurs. Meanwhile, feature inequity shows accelerated growth

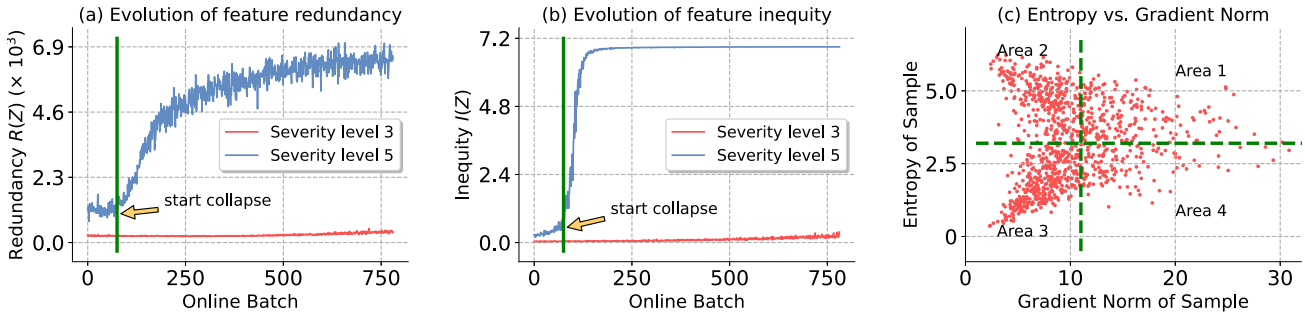


Fig. 3. Failure case analyses (a-b) of online test-time entropy minimization [8]. (a) and (b) illustrate how feature redundancy and inequity evolve during TTA with and without model collapse. Both feature redundancy and feature inequity increase sharply when the model collapses, whereas the stable model (no collapse) keeps them within a low-value range. (c) investigates the relationship between the sample's entropy and the gradient norm, which motivates our sample filtering strategy for removing unreliable gradients, as described in Section V-A. All experiments are conducted on shuffled ImageNet-C of Gaussian noise with ResNet50 (GN), and a larger (severity) level denotes a more severe distribution shift.

during early TTA under severe shift, which reveals an unfolding collapse inside the model.

Inspired by the above insights, we propose to enhance SAR with feature regularization during TTA, namely SAR^2. However, naively applying feature regularization to raw samples from the wild test data stream can harm the TTA performance (*e.g.*, pushing features of the same class apart) or become ill-posed (*e.g.*, using a single sample). To address this, instead of regularizing the features of individual samples, SAR^2 applies regularization to the class centroids, which effectively preserves intra-class compactness while penalizing degraded representations. SAR^2 introduces an exponentially updated feature bank to maintain a centroid per class over the test data stream, and provides a stable reference for the missing classes in the current mini-batch. Feature redundancy and inequity are then applied on class centroids augmented by the feature bank. Extensive experiments demonstrate that both feature redundancy and inequity regularization reduce the risk of collapse, and SAR^2 substantially enhances the stability and efficacy of TTA in the wild. Our contributions are summarized as follows:

- We propose a Sharpness-Aware and Reliable (SAR) optimization scheme for test-time entropy minimization. We analyze and verify empirically that batch-agnostic norm layers (*i.e.*, group norm and layer norm) are more stable than batch norm for test-time adaptation under wild test settings, *i.e.*, mix domain shifts, small test batch sizes, and online imbalanced label distribution shifts (Fig. 1). Based on group/layer norm–equipped models, we further identify that the model collapse of test-time entropy minimization in wild test scenarios is linked to large/noisy gradients. Therefore, we devise SAR to effectively mitigate this, by excluding unreliable samples from TTA and performing sharpness-aware updates to favor flat entropy minima.
- We further extend SAR with feature **r**egularization (namely SAR^2) at test time. SAR^2 introduces two metrics, *i.e.*, feature redundancy and inequity, to provide deeper insights into the unstable learning process of wild TTA, revealing that declines in feature redundancy coincide with effective adaptation, while the accelerated growth in feature inequity foretells a collapse in representation (Fig. 3). Motivated by this, SAR^2 develops a feature redundancy and inequity regularizer for TTA, and further introduces a feature bank and a centroid-based regularization method to stabilize test-time feature regularization in the wild.
- Extensive experiments demonstrate that SAR consistently improves TTA stability under the challenging, wild test settings. Moreover, by explicitly measuring feature drifts and penalizing representation degradation, SAR^2 introduces a more interpretable solution for mitigating collapse and substantially enhances the stability and effectiveness of TTA across various wild scenarios.

II. Related Work

We discuss existing adaptation methods with and without target data, and further review sharpness-aware optimization, neural collapse, and self-supervised feature dimensional collapse works.

Adaptation without Target Data: The problem of conquering distribution shifts has been studied in a number of works at *training time*, including domain generalization [21], [22], [23], increasing the training dataset size [24], various data augmentation techniques [25], [26], [27], [28], [29], to name just a few. These methods aim to pre-anticipate or simulate the possible shifts of test data at training time, so that the training distribution can cover the possible shifts of test data. However, pre-anticipating all possible test shifts at training time may be infeasible and these training strategies are often more computationally expensive. Instead of improving generalization ability at training time, we conquer test shifts by directly learning from the test data.

Adaptation with Target Data: We divide the discussion on related methods that exploit target data into 1) unsupervised domain adaptation (adapt offline) and 2) test-time adaptation (adapt online).

- *Unsupervised domain adaptation (UDA):* Conventional UDA jointly optimizes on the labeled source and unlabeled target data to mitigate distribution shifts, such as devising a domain discriminator to align source and target domains at feature level [30], [31], [32], [33] and aligning the prototypes of source and target domains through a contrastive learning

manner [34]. Recently, *source-free UDA* methods have been proposed to resolve the adaptation problem when source data are absent, such as generative-based methods that generate source images or prototypes from the model [35], [36], [37], and information maximization [38], [39]. These methods adapt models on a whole test set, in which the adaptation is offline and often requires multiple training epochs, and thus are hard to be deployed in online testing scenarios.

- *Test-time adaptation (TTA):* According to whether alter training, TTA methods can be mainly categorized into two groups. i) *Test-Time Training (TTT)* [7] jointly optimizes a source model with both supervised and self-supervised losses, and then conducts self-supervised learning at test time. The self-supervised losses can be rotation prediction [40] in TTT or contrastive-based objectives [41] in TTT++ [11] and MT3 [10], *etc.* ii) *Fully Test-Time Adaptation* [8], [12], [42] does not alter the training process and can be applied to any pre-trained model, including adapting the statistics in batch normalization layers [43], [44], [45], [46], [47], unsupervised entropy minimization [8], [12], [48], prediction consistency maximization [48], [49], [50], *etc.* Nevertheless, prior TTA methods are shown to be unstable in the online adaptation process and are sensitive to insufficient data (small batches), mixed domains, and online imbalanced shifted label distribution (see Fig. 1). Our SAR [51] first analyzes what hinders the stability of TTA under the above practical test settings and proposes associated solutions to stabilize TTA under various wild test scenarios.

Recently, the observations in SAR have been widely noted in the TTA community [52] and SAR has encouraged a series of studies for TTA in the wild [19], [20], [53], [54]. Memory-bank methods [53], [55], [56] introduce a data bank at testing to perform class-balanced sampling under online imbalanced label shifts. In contrast, ROID [20] assumes a uniform target class prior and directly corrects label-imbalance via prior adjustment. Considering the potential risk of model collapse in TTA, RDump [57] periodically resets model parameters during the continuous adaptation process, and TTA-Monitor [58] tracks predictive performance and raises alerts when the model is at risk of collapse. To enhance learning stability, DeYO [19] exploits shape information to select test samples for entropy minimization, while ReCAP [54] further introduces local consistency to ensure entropy minimization favors predictions with large margins to all class boundaries. Nevertheless, prior approaches focus on mitigating collapse through input- or output-level strategies, while leaving the intrinsic causes of collapse underexplored. In this work, we reveal the effects of TTA on model representation through the lens of redundancy and inequity, and further introduce test-time feature regularization as a more robust and interpretable framework to enhance the stability and efficacy of TTA across various test scenarios in the wild.

Sharpness-aware Minimization (SAM): SAM [59] optimizes both a supervised objective (*e.g.*, cross-entropy) and the sharpness of loss surface, aiming to find a flat minimum that has good generalization ability [60]. SAM and its variants [61], [62], [63], [64] have shown outstanding performance on several deep learning benchmarks. In this work, when we analyze the failure reasons of test-time entropy minimization, we find that some noisy samples that produce gradients with large norms harm the adaptation and thus lead to model collapse. To alleviate this, we propose to minimize the sharpness of the test-time entropy loss surface so that the online model update is robust to those noisy/large gradients.

Neural Collapse (NC): NC [65], [66] studies a benign geometric convergence in the terminal phase of supervised training. NC mainly points out: 1) within-class features collapse to their class means, 2) class means form a simplex Equiangular Tight Frame, and 3) classifiers align with these means. Recent NC-TTA [67] further uses this benign NC phenomenon, *i.e.*, feature-classifier alignment, to explain and improve adaptation under out-of-distribution shifts. In contrast, our work studies a different phenomenon: degenerate prediction collapse during online entropy-minimization-based TTA under wild shifts, such as mixed corruptions, small batches, and online label imbalance. Instead of measuring ETF properties or feature-classifier alignment, we introduce feature redundancy and feature inequity to characterize the harmful representation degradation during TTA and motivate our centroid-bank-based regularization.

Dimensional Collapse: in self-supervised learning (SSL) mainly studies whether representations collapse into a low-dimensional subspace during pre-training, usually relying on feature correlation, covariance, or cross-correlation computed from large and relatively stable batch statistics [68], [69], [70]. However, these do not hold in wild online TTA, where test streams can be non-i.i.d., imbalanced, and even limited to batch size 1. Rather than foucusing on pre-training, our work studies a different failure mode at test time: an already trained model may gradually predict most samples as the same class under online entropy minimization. To make feature regularization applicable to this online setting, SAR^2 does not directly decorrelate raw mini-batch features. Instead, it maintains class centroids in an online feature bank and regularizes their feature redundancy, making feature regularization more suitable for small-batch and imbalanced test-time streams. Beyond the correlation statistics commonly considered in SSL, we further introduce a newly designed feature inequity measure to capture whether the learned representation becomes biased toward certain classes, which is essential for monitoring collapse in wild TTA.

III. Preliminaries

We revisit two main categories of test-time adaptation methods in this section for the convenience of further analyses.

Test-time Training (TTT): Let $f_\Theta(\mathbf{x})$ denote a model trained on $\mathcal{D}_{train} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with parameter Θ, where $\mathbf{x}_i \in \mathcal{X}_{train}$ (training data space) and $y_i \in \mathcal{C}$ (label space). The goal of test-time adaptation [7], [8] is to boost $f_\Theta(\mathbf{x})$ on out-of-distribution test samples $\mathcal{D}_{test} = \{\mathbf{x}_j\}_{j=1}^M$, where $\mathbf{x}_j \in \mathcal{X}_{test}$ (testing data space) and $\mathcal{X}_{test} \neq \mathcal{X}_{train}$. Sun et al. [7] first propose the TTT pipeline, in which at **training phase** a model is trained on $\mathcal{D}_{train}$ via both cross-entropy $\mathcal{L}_{CE}$ and self-supervised rotation prediction $\mathcal{L}_S$:

$$\min_{\Theta_b,\Theta_c,\Theta_s} \mathbb{E}_{\mathbf{x}\in\mathcal{D}_{train}}[\mathcal{L}_{CE}(\mathbf{x};\Theta_b,\Theta_c) + \mathcal{L}_S(\mathbf{x};\Theta_b,\Theta_s)], \quad (1)$$

where Θ_b is the task-shared parameters (shadow layers), Θ_c and Θ_s are task-specific parameters (deep layers) for $\mathcal{L}_{CE}$ and $\mathcal{L}_S$, respectively. At **testing phase**, given a test sample $\mathbf{x}$, TTT first updates the model with self-supervised task: $\Theta_b' \leftarrow \arg\min_{\Theta_b} \mathcal{L}_S(\mathbf{x}; \Theta_b, \Theta_s)$ and use the updated model weights Θ_b' to perform final prediction via $f(\mathbf{x}; \Theta_b', \Theta_c)$. Methods like TTT++ [11] and TTT-MAE [71] also explore using contrastive or reconstruction objectives as $\mathcal{L}_S$.

Fully Test-time Adaptation (TTA): The pipeline of TTT needs to alter the original model training process, which may be infeasible when training data is unavailable due to privacy/storage concerns. To avoid this, Tent [8] proposes fully TTA, which provides the flexibility to adapt arbitrary pre-trained models for a given test mini-batch by conducting entropy minimization, formulated as:

$$\min E(\mathbf{x}; \Theta) = -\sum_c f_\Theta(c|\mathbf{x}) \log f_\Theta(c|\mathbf{x}) \tag{2}$$

where c denotes the c-th class. This method removes the need for an auxiliary self-supervised branch and is more efficient and broadly applicable than TTT, as shown in Table II.

IV. What Causes Unstable Test-Time Adaptation Under Wild Scenarios?

Although prior TTA methods have exhibited great potential for out-of-distribution generalization, its success may rely on some underlying test prerequisites (as illustrated in Fig. 1): 1) test samples have the same distribution shift type; 2) adapting with a batch of samples each time, 3) the test label distribution is uniform during the whole online adaptation process, which, however, are easy to be violated in the wild world. In wild scenarios (Fig. 1 ②), prior methods may perform poorly or even fail.

In this section, we seek to analyze the underlying reasons why TTA fails under wild testing scenarios described in Fig. 1 from a unified perspective and then propose associated solutions (c.f. Section V). We start by investigating the impacts of norm layer on TTA stability and further dig into the failure modes of entropy-based methods with batch-agnostic norms, *e.g.*, group norm, from both the output and feature level. Our findings are depicted below.

Batch Normalization Hinders Stable TTA: In TTA, prior methods often conduct adaptation on pre-trained models with batch normalization (BN) layers [13], and most of them are built upon test-time BN statistics calibration [8], [12], [43], [44], [45], [48], [72]. Specifically, for a layer with d-dimensional input $\mathbf{x} = (x^{(1)} \dots x^{(d)})$, the batch normalized output are: $y^{(k)} = \gamma^{(k)} \widehat{x}^{(k)} + \beta^{(k)}$, where $\widehat{x}^{(k)} = (x^{(k)} - \mathrm{E}[x^{(k)}]) / \sqrt{\mathrm{Var}[x^{(k)}]}$. Here, $\gamma^{(k)}$ and $\beta^{(k)}$ are learnable affine parameters. BN adaptation methods calculate mean $\mathrm{E}[x^{(k)}]$ and variance $\mathrm{Var}[x^{(k)}]$ over (a batch of) **test samples.**

However, in wild TTA, all three practical adaptation settings (in Fig. 1) in which TTA may fail will result in problematic mean and variance estimation. **First**, BN statistics indeed represent a distribution and ideally each distribution should have its own statistics. Simply estimating shared BN statistics of multiple distributions from mini-batch test samples unavoidably obtains limited performance, such as in multi-task/domain learning [73]. **Second**, the quality of estimated statistics relies on the batch size, and it is hard to use very few samples (*i.e.*, small batch size) to estimate it accurately. **Third**, the imbalanced label shift will also result in biased BN statistics towards some specific classes in the dataset. Based on the above, we posit that batch-agnostic norm layers, *i.e.*, agnostic to the way samples are grouped into a batch, are more suitable for performing TTA, such as group norm (GN) [14] and layer norm (LN) [15], and devise our method based on GN/LN models in Section V. To verify the above claim, we empirically investigate the effects of different normalization layers (including BN, GN, and LN) in TTA (including TTT and Tent) in Section VI. From the results, we observe that models equipped with GN and LN are more stable than models with BN when performing online test-time adaptation under three practical test settings (in Fig. 1) and have fewer failure cases. The detailed empirical studies are put in Section VI for the coherence of presentation.

Online Entropy Minimization Tends to Result in Collapsed Trivial Solutions,i.e., Predict All Samples to the Same Class: Although TTA performs more stable on GN and LN models, it does not always succeed and still faces several failure cases. For example, entropy minimization (Tent) on GN models (ResNet50-GN) tends to collapse, especially when the distribution shift extent is severe. In this paper, we aim to stabilize online fully TTA under various practical test settings. To this end, we first analyze the failure reasons, in which we find models are often optimized to collapse trivial solutions. We illustrate this issue in the following.

During the online adaptation process, we record the predicted class and the gradients norm (produced by entropy loss) of ResNet50-GN on shuffled ImageNet-C of Gaussian noise. By comparing Fig. 2(a) and (b), entropy minimization is shown to be unstable and may occur collapse when the distribution shift is severe (*i.e.*, severity level 5). From Fig. 2(a), as the adaptation goes by, the model tends to predict all input samples to the same class, even though these samples have different ground-truth classes, called model collapse. Meanwhile, we notice that along with the model starts to collapse the ℓ_2-norm of gradients of all trainable parameters increases significantly and then degrades to almost 0 (as shown in Fig. 2(c)), while on severity level 3 the model works well and the gradients norm keep in a stable range all the time. This indicates that some test samples produce large gradients that may hurt the adaptation and lead to model collapse.

Feature Redundancy and Inequity are Model Collapsing Signals: Our previous study reveals that TTA may push the network into a trivial solution, yet the underlying mechanisms remain under-explored. As shown in Fig. 2(c), we notice a continual increase in the gradient norm before model collapse, implying collapse as a cumulative consequence which may be tractable. To understand how online TTA gradually affects a model and when it leads to successful or failed adaptations, we further dig into the shifts of model behaviors at the internal *feature level*. Specifically, we monitor two feature-level statistics throughout TTA:1) **Redundancy**, which measures the similarity between each dimension of model representation, where highly correlated dimensions increase the estimated redundancy; and 2) **Inequity**, which quantifies the extent of model bias towards

predicting some particular classes, measured by the prediction entropy of a centroid computed from features across different classes. The intuition is that if each class contributes fairly to the classification task, the centroid of features from all classes should be classified with maximum uncertainty—meaning the classifier would assign nearly uniform probabilities to all classes for this centroid. Formally, Let $z_b = g_\Theta(x_b) \in \mathbb{R}^D$ and $\mathbf{Z} = [z_1, \ldots, z_B]$ denote the features of a batch, and $\mu_B = \frac{1}{B}\sum_b z_b$ be the centroid of the batch feature. Feature redundancy $R(\cdot)$ and inequity $I(\cdot)$ are then given by:

$$R(\mathbf{Z}) = \frac{1}{D-1}\sum_{i\neq j}\mathcal{C}_{ij}^2, \quad \mathcal{C} = \frac{1}{B}\cdot\frac{(\mathbf{Z}-\mu_B)^\top(\mathbf{Z}-\mu_B)}{\sigma_B(\mathbf{Z})^2}; \tag{3}$$

$$I(\mathbf{Z}) = \log C + \sum_c p'_c \log p'_c, \quad p' = \mathrm{softmax}\big(h(\mu_B)\big), \tag{4}$$

where $\sigma_B(\cdot)$ is the standard variance, $\mathcal{C}$ is the covariance matrix, C is the number of classes, and $h(\cdot)$ is the classifier head.

We conduct experiments on the ResNet50-GN model following Fig. 2 and (3) & (4) are calculated over the current test batch. Our results in Fig. 3 reveal two key observations: 1) *Shift severity determines the initial health of the representation.* A more severe distribution shift (*i.e.*, severity level 5) induces larger redundancy and inequity in model representation before adaptation takes place, as in the first points of Fig. 3(a)–(b). Appendix D, available online further confirms such monotonic increment across all five severity levels, providing a new understanding of how distribution shift affects a model's performance. 2) *The changes in feature redundancy and inequity foretell TTA effectiveness and collapse.* During TTA on severity level 5, feature redundancy $R(\mathbf{Z})$ first remains stable (or decreases on VitBase, see Appendix D.6, available online) when TTA still improves accuracy, followed by a significant increase that positions model collapse. Meanwhile, feature inequity $I(\mathbf{Z})$ shows accelerated growth during early adaptation, which then results in model collapse and a further explosion in feature inequity. In contrast, when TTA works well under severity level 3, feature inequity and redundancy remain within a stable range. This underscores a strong correlation between feature inequity/redundancy and model collapse, offering new insights to improve TTA stability and effectiveness in the wild.

V. Stable and Effective Test-Time Adaptation in Dynamic Wild World

Based on the above analyses, we aim to resolve model collapse during TTA in the wild from both the model's output level and the feature level: At the output level, we first select reliable test samples with small entropy values to reduce large and noisy gradients, which are then used for sharpness-aware entropy minimization (*namely* SAR, c.f. Section V-A); At the feature level, we introduce a prototype feature bank to enable test-time feature regularization for TTA in the wild settings, where we apply both redundancy and inequity regularization along with entropy minimization to mitigate feature collapse and enhance the TTA effectiveness (*namely* SAR^2, c.f. Section V-B).

A. Sharpness-Aware Reliable Entropy Minimization

Based on the above analyses, two most straightforward solutions to mitigate model collapse during entropy minimization are filtering out test samples according to the sample gradients or performing gradients clipping. However, these are not feasible since the gradients norms for different models and distribution shift types have different scales, and thus it is hard to devise a general method to set the threshold for sample filtering or gradient clipping (see Section VII-B for more analyses). We propose our solutions as follows.

Reliable Entropy Minimization: Since directly filtering samples with gradients norm is infeasible, we first investigate the relation between entropy loss and gradients norm and seek to remove samples with large gradients based on their entropy. Here, the entropy depends on the model's output class number C and it belongs to $(0, \ln C)$ for different models and data. In this sense, the threshold for filtering samples with entropy is easier to select. As shown in Fig. 3(c), selecting samples with small loss values can remove part of samples that have large gradients (`area@1`) out of adaptation. Formally, let $E(\mathbf{x};\Theta)$ be the entropy of sample $\mathbf{x}$, the selective entropy minimization is defined by:

$$\min_\Theta S(\mathbf{x})E(\mathbf{x};\Theta), \text{ where } S(\mathbf{x}) \triangleq \mathbb{I}_{\{E(\mathbf{x};\Theta)<E_0\}}(\mathbf{x}). \tag{5}$$

Here, Θ denote model parameters, $\mathbb{I}_{\{\cdot\}}(\cdot)$ is an indicator function and E_0 is a pre-defined parameter. Note that the above criteria will also remove samples within `area@2` in Fig. 3(c), in which the samples have low confidence and thus are unreliable [12].

Sharpness-aware Entropy Minimization: Through (5), we have removed test samples in `area@1&2` in Fig. 3(c) from adaptation. Ideally, we expect to optimize the model via samples only in `area@3`, since samples in `area@4` still have large gradients and may harm the adaptation. However, it is hard to further remove the samples in `area@4` via a filtering scheme. Alternatively, we seek to make the model insensitive to the large gradients contributed by samples in `area@4`. Here, we encourage the model to go to a flat area of the entropy loss surface. The reason is that a flat minimum has good generalization ability and is robust to noisy/large gradients, *i.e.*, the noisy/large updates over the flat minimum would not significantly affect the original model loss, while a sharp minimum would. To this end, we jointly minimize the entropy and the sharpness of entropy loss by:

$$\min_\Theta E^{SA}(\mathbf{x};\Theta), \quad \text{where} \quad E^{SA}(\mathbf{x};\Theta) \triangleq \max_{\|\boldsymbol{\epsilon}\|_2\leq\rho} E(\mathbf{x};\Theta+\boldsymbol{\epsilon}). \tag{6}$$

Here, the inner optimization seeks to find a weight perturbation $\boldsymbol{\epsilon}$ in a Euclidean ball with radius ρ that maximizes the entropy. The sharpness is quantified by the maximal change of entropy between Θ and $\Theta+\boldsymbol{\epsilon}$. This bi-level problem encourages the optimization to find flat minima. To address problem (6), we follow SAM [59] that first approximately solves the inner optimization via first-order Taylor expansion, *i.e.*,

$$\boldsymbol{\epsilon}^*_E(\mathbf{x};\Theta) \triangleq \arg\max_{\|\boldsymbol{\epsilon}\|_2\leq\rho} E(\mathbf{x};\Theta+\boldsymbol{\epsilon})$$

$$\approx \arg\max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} E(\mathbf{x};\Theta) + \boldsymbol{\epsilon}^T \nabla_\Theta E(\mathbf{x};\Theta)$$

$$= \arg\max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} \boldsymbol{\epsilon}^T \nabla_\Theta E(\mathbf{x};\Theta). \tag{7}$$

Then, $\hat{\boldsymbol{\epsilon}}_E(\mathbf{x};\Theta)$ that solves this approximation is given by the solution to a classical dual norm problem:

$$\hat{\boldsymbol{\epsilon}}_E(\mathbf{x};\Theta) = \rho \nabla_\Theta E(\mathbf{x};\Theta) / \left\|\nabla_\Theta E(\mathbf{x};\Theta)\right\|_2 . \tag{8}$$

Substituting $\hat{\boldsymbol{\epsilon}}(\Theta)$ back into (6) and differentiating, by omitting the second-order terms for computation acceleration, the final gradient approximation is:

$$\nabla_\Theta E^{SA}(\mathbf{x};\Theta) \approx \nabla_\Theta E(\mathbf{x};\Theta)\big|_{\Theta+\hat{\boldsymbol{\epsilon}}_E(\mathbf{x};\Theta)} . \tag{9}$$

Then, our sharpness-aware reliable entropy minimization for test-time model update (*termed* SAR) is given by:

$$\min_{\tilde{\Theta}} S(\mathbf{x}) E^{SA}(\mathbf{x};\Theta). \tag{10}$$

Here, $\tilde{\Theta} \subset \Theta$ denote learnable parameters. In addition, to handle extremely hard cases where SAR may still fail, we also introduce a Model Recovery Scheme to guarantee a minimum level of stability. We maintain a moving average e_m of entropy loss values and reset $\tilde{\Theta}$ to be the original once e_m falls below a small threshold e_0, since collapsed models typically produce abnormally (very) small entropy losses. this mechanism is rarely triggered by SAR and never triggered by SAR^2, as in Section VII-A.

B. Test-Time Feature Regularization in the Wild

Our analysis in Section IV reveals that model representations tend to become redundant and unequal (*i.e.*, bias towards certain classes) during a collapsed TTA process. To address this, a simple solution would be to use feature regularization along with TTA. Methods such as VicReg [74] and Barlow Twins [70] explore using the feature redundancy regularizer in self-supervised learning. Nevertheless, their effectiveness relies on the assumption of i.i.d. data and a large batch size. In online TTA, this assumption generally fails since: *C1*) test data can be highly correlated and produce non-i.i.d. data stream, *i.e.*, the online imbalanced label shift scenario; and *C2*) real-world TTA adapts on micro batches or even a single sample at a time. In this case, directly applying the feature regularizer on test data may harm the TTA performance (*e.g.*, enforcing dissimilar features among the same class) or become ill-posed (*e.g.*, using a single sample), and effective feature regularization at test time remains challenging.

Feature Bank for Online Representation Regularization under Wild TTA: To address the above pitfalls, we introduce two simple but necessary modifications: 1) We reformulate feature regularization to operate on class centroids instead of on individual samples. This reformulation mitigates collapse by encouraging full utilization of the feature space across classes, while bypassing the risks of pushing the features of the same class apart, *i.e.*, efficiently addressing challenge *C1*. 2) We maintain an exponentially updated feature bank that stores one centroid per class over the test data stream. This bank provides a stable reference for the missing classes even when the test batch contains a single sample, *i.e.*, tackling challenge *C2*. Formally, let $\mathcal{B}^t$ be a mini-batch, $g_\Theta(\mathbf{x}) \in \mathbb{R}^D$ be the extracted feature of sample $\mathbf{x}$, and $\hat{y}(\mathbf{x})$ denote the predicted label, we compute a centroid $\mathbf{c}_i$ for every class that *appears* in the mini-batch as:

$$\mathbf{c}_i = \frac{\sum_{\mathbf{x}\in\mathcal{B}} \mathbf{1}\big[\hat{y}(\mathbf{x}) = i\big] \cdot g_\Theta(\mathbf{x})}{\sum_{\mathbf{x}\in\mathcal{B}} \mathbf{1}\big[\hat{y}(\mathbf{x}) = i\big]}, \quad i \in \hat{\mathcal{Y}}_{\mathcal{B}^t}. \tag{11}$$

where $\hat{\mathcal{Y}}_{\mathcal{B}^t}$ is the set of pseudo-labeled classes present in $\mathcal{B}^t$. Meanwhile, for classes *absent* from $\mathcal{B}^t$, we refer class feature from the feature bank $\tilde{\mathbf{C}}=\{\tilde{c}_i\}_{i=1}^C$, and the feature matrix $\mathbf{C}^t$ of a mini-batch $\mathcal{B}^t$ for stable test-time feature regularization becomes:

$$\mathbf{C}_i^t = \begin{cases} \mathbf{c}_i, & i \in \hat{\mathcal{Y}}_{\mathcal{B}^t}, \\ \tilde{\mathbf{c}}_i, & \text{otherwise}, \end{cases} \tag{12}$$

The bank is initialized as empty, and class centroids are inserted or updated into the feature bank via the exponential moving average:

$$\tilde{\mathbf{c}}_i' = (1-\lambda)\,\tilde{\mathbf{c}}_i + \lambda\,\mathbf{c}_i, \;\; i \in \hat{\mathcal{Y}}_{\mathcal{B}^t}, \tag{13}$$

while leaving classes that absent from $\mathcal{B}^t$ unchanged.

Feature Redundancy and Inequity Regularizer: Given the feature matrix $\mathbf{C}^t$ derived from (12), we can then perform regularization under small test batches (even a single sample) or imbalanced test data, *i.e.*, the wild settings. Specifically, we calculate the feature redundancy $R(\mathbf{C}^t)$ and inequity $I(\mathbf{C}^t)$ from matrix $\mathbf{C}^t$ as test-time regularizers, following (3) & (4).

During TTA, we minimize $I(\mathbf{C}^t)$ to promote unbiased representations, alleviating the potential risk of collapse induced by entropy minimization. We also reduce $R(\mathbf{C}^t)$ to encourage non-redundant representations of test samples for enhancing discrimination ability, which improves generalization against

Algorithm 1: The pipeline of proposed SAR and SAR^2.

Require Test samples $\mathcal{D}_{test}=\{\mathbf{x}_j\}_{j=1}^M$, model $f_\Theta(\cdot)$ with its feature extractor $g(\cdot)$ and classifier $h(\cdot)$, trainable parameters $\tilde{\Theta} \subset \Theta$, feature bank $\tilde{\mathbf{C}}$, feature bank warm-up threshold ζ.

Initialize $\tilde{\Theta}_0 = \tilde{\Theta}$, feature bank $\tilde{\mathbf{C}} = \varnothing$.
for a batch $\mathcal{B}^t=\{\mathbf{x}_j\}_{j=1}^B$ in $\mathcal{D}_{test}$ **do**
  Compute features $\mathbf{z}_j=g(\mathbf{x}_j)$ and predictions $\hat{y}_j=h(\mathbf{z}_j)$.
  `For SAR:`
  Calculate entropy loss $E(\mathbf{x}_j;\Theta)$ via (2).
  Update model ($\tilde{\Theta} \subset \Theta$) with (10).
  `For SAR`^{2}`:`
  Retrieve augmented features $\mathbf{C}^t$ via (12).
  if $\mathbf{C}^t$ contains more than ζ features **then**
    Calculate entropy loss $E(\mathbf{x}_j;\Theta)$ via (2).
    Compute feature redundancy $R(\mathbf{C}^t;\Theta)$ via (3).
    Compute feature inequity $I(\mathbf{C}^t;\Theta)$ using (4).
    Update model ($\tilde{\Theta} \subset \Theta$) with (16).
  end if
  Refresh the feature bank $\tilde{\mathbf{C}}$ via (13).
end for

Ensure: The predictions $\{\hat{y}_j\}_{j=1}^M$ for all $\mathbf{x} \in \mathcal{D}_{test}$.

distribution shifts even when used alone during TTA, as shown in Appendix D.8, available online. In SAR^2, these regularizers are also combined with our sharpness-aware optimization mechanism for stable updates. Following (8), let $\hat{\boldsymbol{\epsilon}}_R(\mathbf{C}^t;\Theta)$ and $\hat{\boldsymbol{\epsilon}}_I(\mathbf{C}^t;\Theta)$ be the worst-case perturbations, we approximate the gradients of the sharpness-aware feature redundancy $R^{SA}(\mathbf{C}^t;\Theta)$ and inequity $I^{SA}(\mathbf{C}^t;\Theta)$ regularizer as:

$$\nabla_\Theta R^{SA}(\mathbf{C}^t;\Theta) \approx \nabla_\Theta R^{SA}(\mathbf{C}^t;\Theta)\big|_{\Theta+\hat{\boldsymbol{\epsilon}}_R(\mathbf{C}^t;\Theta)}, \tag{14}$$

$$\nabla_\Theta I^{SA}(\mathbf{C}^t;\Theta) \approx \nabla_\Theta I^{SA}(\mathbf{C}^t;\Theta)\big|_{\Theta+\hat{\boldsymbol{\epsilon}}_I(\mathbf{C}^t;\Theta)}. \tag{15}$$

Discussion: It is worth noting that inequity $I(\cdot)$ operates at the *prototype feature centroid* by maximizing the entropy of the centroid prediction, penalizing biased model representation towards a specific class. In contrast, redundancy $R(\cdot)$ is *centroid-invariant* and explicitly targets inter-dimension correlations: it is computed on mean-centered features (via centroid subtraction) to penalize statistical dependencies among feature dimensions, thereby encouraging a decorrelated representation independent of the centroid. Together, they exert complementary effects: inequity ensures well-conditioned centroids by penalizing class bias, while redundancy enforces decorrelated features through centroid-invariant regularization. As shown in Table VII, each regularizer individually improves TTA efficacy under wild test settings, and their combination yields a significant further boost.

Overall Optimization: In summary, our sharpness-aware reliable entropy minimization with feature regularization (*termed* SAR^2) is given by:

$$\min_{\tilde{\Theta}} S(\mathbf{x})E^{SA}(\mathbf{x};\Theta) + \alpha R^{SA}(\mathbf{C}^t) + \beta I^{SA}(\mathbf{C}^t), \tag{16}$$

where $S(\mathbf{x})$ and $E^{SA}(\mathbf{x};\Theta)$ are defined in (5) and (6) respectively, α and β are balancing coefficients, and $\tilde{\Theta} \subset \Theta$ denote learnable parameters during TTA. In addition, to prevent unstable optimization during the warm-up phase of the feature bank, we perform (16) only when $\mathbf{C}^t$ contains no less than ζ non-empty class features. We summarize our methods in Algorithm 1.

VI. Empirical Studies of Normalization Layer Effects in TTA

This section designs experiments to illustrate how test-time adaptation (TTA) performs on models with different norm layers (including BN, GN and LN) under wild test settings described in Fig. 1. We verify two representative methods introduced in Section III, *i.e.*, self-supervised **TTT** [7] and unsupervised **Tent** (a fully TTA method) [8]. Considering that the norm layers are often coupled with mainstream network architectures, we conduct adaptation on ResNet-50-BN (R-50-BN), ResNet-50-GN (R-50-GN) and VitBase-LN (Vit-LN). All adopted model weights are public available and obtained from `torchvision` or `timm` repository [75]. Implementation details of experiments in this section can be found in Appendix A.2, available online.

1) Norm Layer Effects in TTA Under Small Test Batch Sizes: We evaluate TTA methods (TTT and Tent) with different batch sizes (BS), selected from {1, 2, 4, 8, 16, 32, 64}. Due to GPU memory limits, we only report results of BS up to 8 or 16 for TTT (in original TTT BS is 1), since TTT needs to augment each test sample multiple times (for which we set to 20 by following [12]).

From Fig. 4, we have: **i)** For Tent, compared with R-50-BN, R-50-GN and Vit-LN are less sensitive to small test batch sizes. The adaptation performance of R-50-BN degrades severely when the batch size goes small ($<$8), while R-50-GN/Vit-LN show stable performance across various batch sizes (Vit-LN on levels 5&3 and R-50-GN on level 3, in subfigures (b)&(d)). It is worth noting that Tent with R-50-GN and Vit-LN not always succeeds and also has failure cases, such as R-50-GN on level 5 (Tent performs worse than no adapt), which is analyzed in Section IV. **ii)** For TTT, all R-50-BN/GN and Vit-LN can perform well under various batch sizes. However, TTT with Vit-LN is very unstable and has a large variance over different runs, showing that TTT+VitBase is very sensitive to different sample orders. Here, TTT performs well with R-50-BN under BS 1 is mainly benefited from TTT applying multiple data augmentations to a single sample to form a mini-batch.

2) Norm Layer Effects in TTA Under Mixed Distribution Shifts: We evaluate TTA methods on models with different norm layers when test data come from multiple shifted domains simultaneously. We compare 'no adapt', 'avg. adapt' (average accuracy of adapting on each domain separately) and 'mix adapt' (adapting on mixed and shifted domains) accuracy on ImageNet-C consisting of 15 corruption types. The larger accuracy gap between 'mix adapt' and 'avg. adapt' indicates the more sensitive to mixed domain shifts.

From Fig. 5 we have: **i)** For both Tent and TTT, R-50-GN and Vit-LN perform more stable than R-50-BN under mix domain shifts. Specifically, the mix adapt accuracy of R-50-BN is consistently poor than the average adapt accuracy across different severity levels (in all subfigures (a-d)). In contrast, R-50-GN and Vit-LN are able to achieve comparable accuracy of mix and average adapt, *i.e.*, TTT on R-50-GN (levels 5&3) and Tent on Vit-LN (level 3). **ii)** For R-50-GN and Vit-LN, TTT performs more stable than Tent. To be specific, Tent gets 3/4 failure cases (R-50-GN on levels 5&3, Vit-LN on level 5), which is more than that of TTT. **iii)** The same as Section VI (1), TTT on Vit-LN has large variances over multiple runs, showing TTT+Vit-LN is sensitive to different sample orders.

3) Norm Layer Effects in TTA Under Online Imbalanced Label Shifts: As in Fig. 1(c), during the online adaptation process, the label distribution $Q_t(y)$ at different time-steps t may be different (online shift) and imbalanced. To evaluate this, we first simulate this imbalanced label distribution shift by adjusting the order of input samples (from a test set) as follows.

Online Imbalanced Label Distribution Shift Simulation: Assuming that we have totally T time-steps and T equals to the class number C. We set the probability vector $Q_t(y) = [q_1, q_2, \ldots, q_C]$, where $q_c = q_{max}$ if $c = t$ and $q_c = q_{min} \triangleq (1 - q_{max})/(C-1)$ if $c \neq t$. Here, q_{max}/q_{min} denotes the imbalance ratio. Then, at each $t \in \{1, 2, \ldots, T{=}C\}$, we sample M images from the test set according to $Q_t(y)$. Based on ImageNet-C (Gaussian noise), we construct a new testing set that has online imbalanced label distribution shifts with totally $100(M) \times 1000(T)$ images. Note that we pre-shuffle the class orders in ImageNet-C, since we cannot know which class will come in practice.

From Fig. 6, we have: **i)** For Tent, R-50-GN and Vit-LN are less sensitive than R-50-BN to online imbalanced label

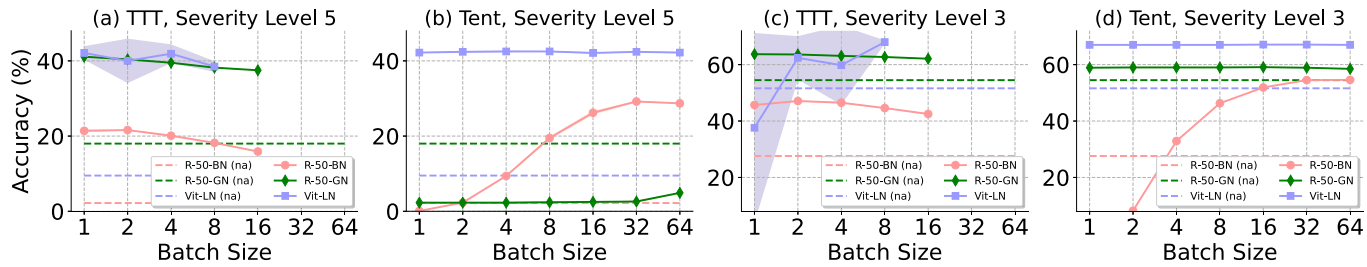


Fig. 4. Batch size effects of different TTA methods under different models (different normalization layers). Experiments are conducted on ImageNet-C of Gaussian noise. We report mean and standard deviation of 3 runs with different random seeds. 'na' denotes no adapt accuracy. Note that except for Vit-LN, the standard deviation is too small to display in the figures.

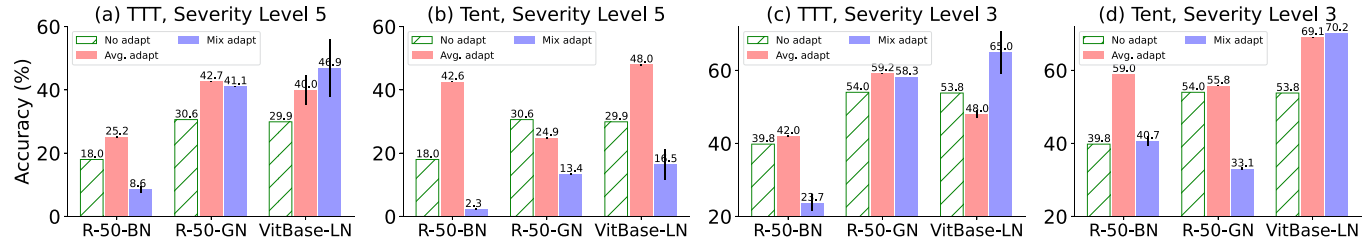


Fig. 5. Performance of TTA methods on different models (different norm layers) under the mixture of 15 different corruption types (ImageNet-C). We report mean&stdev. over 3 independent runs.

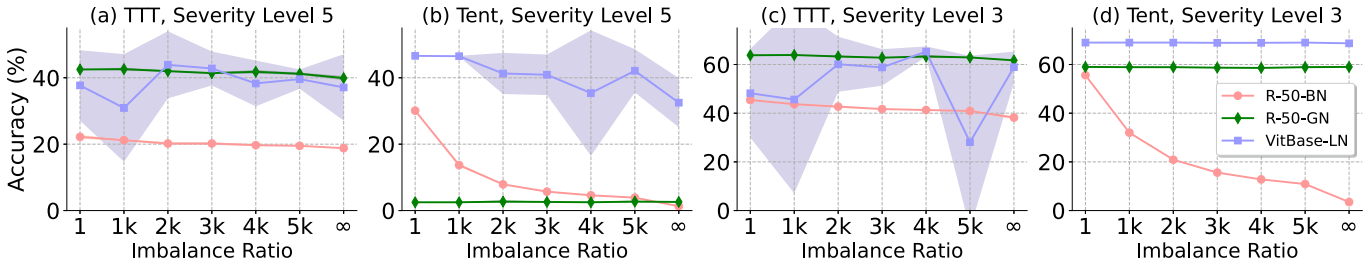


Fig. 6. Performance of TTA methods with different models (different norm layers) under online imbalanced label distribution shifts on ImageNet-C (Gaussian noise). We report mean&stdev. results of 3 runs. Note that except for VitBase-LN, the stdev. is too small to display in the figures.

distribution shifts (see subfigures (b)&(d)). Specifically, the adaptation accuracy of R-50-BN (levels 5&3) degrades severely as the imbalance ratio increases. In contrast, R-50-GN and Vit-LN have the potential to perform stably under various imbalance ratios (*e.g.*, R-50-GN and Vit-LN on level 3). **ii)** For TTT, all R-50-BN/GN and Vit-LN perform relatively stable under label shifts, except for TTT+Vit-LN has large variances. The adaptation accuracy will also degrade but not very severe as the imbalance ratio increases. **iii)** Tent with GN is more sensitive to the extent of distribution shift than BN. Specifically, for imbalanced ratio 1 (all $Q_t(y)$ are uniform) and severity level 5, Tent+R-50-GN fails and performs poorer than no adapt, while Tent+R-50-BN works well.

4) Overall Observations: Based on all the above results, we have: **i)** R-50-GN and Vit-LN are more stable than R-50-BN when performing TTA under wild test settings (see Fig. 1). However, they do not always succeed and still suffer from several failure cases. **ii)** R-50-GN is more suitable for self-supervised TTT than Vit-LN, since TTT+Vit-LN is sensitive to different sample orders and has large variances over different runs. **iii)** Vit-LN is more suitable for unsupervised Tent than R-50-GN, since Tent+R-50-GN is easily to collapse, especially when the distribution shift is severe.

VII. Comparison With State-of-The-Arts

Dataset and Methods: For image classification, we conduct experiments based on ImageNet-C [4], a large-scale and widely used benchmark for out-of-distribution generalization. It contains 15 types of 4 main categories (noise, blur, weather, digital) corrupted images and each type has 5 severity levels. For natural language reasoning, we assess logical reasoning capabilities on Math-500 [77], CollegeMath [78], AIME24, and Minerva [79], ranging from high-school mathematics to advanced competition problems. We compare with the following state-of-the-art methods. DDA [76] performs input adaptation at test time via a diffusion model. Tent [8] minimizes prediction entropy for fully test-time adaptation. MEMO [48] minimizes marginal entropy over different augmented copies w.r.t. a given test sample. EATA [12], ROID [20], DeYO [19], and ReCAP [54] are

TABLE I

COMPARISONS WITH STATE-OF-THE-ART METHODS ON IMAGENET-C (SEVERITY LEVEL 5) UNDER **ONLINE IMBALANCED LABEL SHIFTS**(IMBALANCE RATIO $=\infty$) REGARDING **ACCURACY (%)**. "BN"/"GN"/"LN" IS SHORT FOR BATCH/GROUP/LAYER NORMALIZATION. THE **BOLD** NUMBER INDICATES THE BEST RESULT.

	Noise			Blur				Weather				Digital				
Model+Method	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	Avg.
ResNet50 (BN)	2.2	2.9	1.8	17.8	9.8	14.5	22.5	16.8	23.4	24.6	59.0	5.5	17.1	20.7	31.6	18.0
• MEMO	7.4	8.6	8.9	19.8	13.2	20.8	27.5	25.6	28.6	32.3	60.8	11.0	23.8	33.2	37.7	24.0
• DDA	32.2	33.1	32.0	14.6	16.4	16.6	24.4	20.0	25.5	17.2	52.2	3.2	35.7	41.8	45.4	27.2
• Tent	1.2	1.4	1.4	1.0	0.9	1.2	2.6	1.7	1.8	3.6	5.0	0.5	2.6	3.2	3.1	2.1
• EATA	0.3	0.3	0.3	0.2	0.2	0.5	0.9	0.8	0.9	1.8	3.5	0.2	0.8	1.2	0.9	0.9
ResNet50 (GN)	17.9	19.9	17.9	19.7	11.3	21.3	24.9	40.4	47.4	33.6	69.2	36.3	18.7	28.4	52.2	30.6
• MEMO	18.4	20.6	18.4	17.1	12.7	21.8	26.9	40.7	46.9	34.8	69.6	36.4	19.2	32.2	53.4	31.3
• DDA	42.5	43.4	42.3	16.5	19.4	21.9	26.1	35.8	40.2	13.7	61.3	25.2	37.3	46.9	54.3	35.1
• Tent	2.6	3.3	2.7	13.9	7.9	19.5	17.0	16.5	21.9	1.8	70.5	42.2	6.6	49.4	53.7	22.0
• EATA	27.0	28.3	28.1	14.9	17.1	24.4	25.3	32.2	32.0	39.8	66.7	33.6	24.5	41.9	38.4	31.6
• ROID	32.3	35.0	32.5	15.9	19.8	27.3	30.9	48.0	44.6	52.8	72.2	46.0	34.2	46.9	54.6	39.5
• DeYO	42.5	44.9	43.8	22.2	16.3	41.0	13.2	52.2	51.5	39.7	73.4	52.6	46.9	59.3	59.3	43.9
• ReCAP	42.0	44.1	42.7	19.8	24.3	39.7	40.2	46.0	52.2	57.3	73.1	52.4	33.7	59.4	59.5	45.8
• SAR (ours)	33.1$_{\pm 1.0}$	36.5$_{\pm 0.4}$	35.5$_{\pm 1.1}$	19.2$_{\pm 0.4}$	19.5$_{\pm 1.2}$	33.3$_{\pm 0.5}$	27.7$_{\pm 4.0}$	23.9$_{\pm 5.1}$	45.3$_{\pm 0.4}$	50.1$_{\pm 1.0}$	71.9$_{\pm 0.1}$	46.7$_{\pm 0.2}$	7.1$_{\pm 1.8}$	52.1$_{\pm 0.5}$	56.3$_{\pm 0.1}$	37.2$_{\pm 0.6}$
• SAR^2 (ours)	**43.0**$_{\pm 0.3}$	**45.8**$_{\pm 0.5}$	**43.9**$_{\pm 0.5}$	**35.5**$_{\pm 0.6}$	**36.5**$_{\pm 0.5}$	**44.5**$_{\pm 0.2}$	**48.7**$_{\pm 0.2}$	**56.1**$_{\pm 0.3}$	**55.0**$_{\pm 0.2}$	**61.0**$_{\pm 0.3}$	**73.4**$_{\pm 0.1}$	**55.3**$_{\pm 0.1}$	**54.0**$_{\pm 0.2}$	**61.9**$_{\pm 0.1}$	**61.0**$_{\pm 0.2}$	**51.7**$_{\pm 0.1}$
VitBase (LN)	9.4	6.7	8.3	29.1	23.4	34.0	27.0	15.8	26.3	47.4	54.7	43.9	30.5	44.5	47.6	29.9
• MEMO	21.6	17.4	20.6	37.1	29.6	40.6	34.4	25.0	34.8	55.2	65.0	54.9	37.4	55.5	57.7	39.1
• DDA	41.3	41.3	40.6	24.6	27.4	30.7	26.9	18.2	27.7	34.8	50.0	32.3	42.2	52.5	52.7	36.2
• Tent	32.7	1.4	34.6	54.4	52.3	58.2	52.2	7.7	12.0	69.3	76.1	66.1	56.7	69.4	66.4	47.3
• EATA	35.9	34.6	36.7	45.3	47.2	49.3	47.7	56.5	55.4	62.2	72.2	21.7	56.2	64.7	63.7	49.9
• ROID	44.9	44.7	45.3	48.6	48.9	54.1	50.5	59.9	57.4	67.0	75.2	32.5	60.3	67.5	64.6	54.8
• DeYO	53.5	36.0	54.6	57.6	58.7	63.7	46.2	67.6	66.0	73.2	77.9	66.7	69.0	73.5	70.3	62.3
• ReCAP	53.1	38.5	49.6	57.3	59.0	**63.8**	60.7	67.8	**66.3**	72.9	77.7	66.8	68.2	73.0	70.0	63.0
• SAR (ours)	46.5$_{\pm 3.0}$	43.1$_{\pm 7.4}$	48.9$_{\pm 0.4}$	55.3$_{\pm 0.1}$	54.3$_{\pm 0.2}$	58.9$_{\pm 0.1}$	54.8$_{\pm 0.2}$	53.6$_{\pm 7.1}$	46.2$_{\pm 3.5}$	69.7$_{\pm 0.3}$	76.2$_{\pm 0.1}$	66.2$_{\pm 0.3}$	60.9$_{\pm 0.3}$	69.6$_{\pm 0.1}$	66.6$_{\pm 0.1}$	58.0$_{\pm 0.5}$
• SAR^2 (ours)	**53.7**$_{\pm 0.1}$	**54.5**$_{\pm 0.3}$	**55.0**$_{\pm 0.3}$	**57.8**$_{\pm 0.2}$	**59.0**$_{\pm 0.1}$	63.6$_{\pm 0.2}$	**61.7**$_{\pm 0.1}$	**67.8**$_{\pm 0.2}$	66.0$_{\pm 0.1}$	**73.3**$_{\pm 0.1}$	**77.8**$_{\pm 0.1}$	**67.4**$_{\pm 0.1}$	**68.7**$_{\pm 0.2}$	**73.4**$_{\pm 0.2}$	**70.1**$_{\pm 0.1}$	**64.6**$_{\pm 0.1}$

TABLE II

CHARACTERISTICS OF STATE-OF-THE-ART METHODS. WE EVALUATE THE EFFICIENCY OF DIFFERENT METHODS WITH RESNET50 (GN) ON IMAGENET-C (GAUSSIAN NOISE, LEVEL 5), WHICH CONSISTS OF 50,000 IMAGES. THE REAL RUN TIME IS TESTED VIA A SINGLE V100 GPU. DDA [76] PRE-TRAINS AN ADDITIONAL DIFFUSION MODEL AND THEN PERFORMS INPUT ADAPTATION/DIFFUSION AT TEST TIME. SAR^{2†} IS A VARIANT THAT REMOVES SAM FROM SAR^2. SAR^{2†} INCURS A SLIGHT ACCURACY DROP COMPARED WITH SAR^2 (0.2% UNDER MIXED SHIFTS AND 1.5% UNDER ONLINE LABEL SHIFTS ON IMAGENET-C).

Method	#Forward	#Backward	Other computation	GPU time (s)
MEMO [48]	50,000×65	50,000×64	AugMix [26]	55,980
DDA [76]	50,000×2	0	50,000 diffusion	146,220
TTT [7]	50,000×21	50,000×20	rotation aug.	3,600
Tent [8]	50,000	50,000	n/a	110
EATA [12]	50,000	26,196	regularizer	114
SAR (ours)	50,000 + 12,710	12,710×2	n/a	115
SAR^2 (ours)	50,000×2	50,000×2	feature bank	226
SAR^{2†} (ours)	50,000	50,000	feature bank	114

entropy-based methods with active sample selection for online fully TTA. TLM [80] refines the prediction entropy on input sequences.

Models and Implementation Details: We conduct experiments on ResNet50-BN/GN and VitBase-LN that are obtained from `torchvision` or `timm` [75]. For both SAR and SAR^2, we use SGD as the update rule, with a momentum of 0.9 and a batch size of 64 (except for the experiments of batch size=1). Learning rates are 0.00025/0.001 in SAR, and 0.001/0.005 in SAR^2 for ResNet/Vit models, respectively. The threshold E_0 in (5) is set to $0.4\times \ln 1000$ per EATA [12]. ρ in (6) is set by the default value 0.05 in [59]. The coefficients α and β in (16) are set to $10^3/D$ and 50, respectively, where D is the feature dimension. ζ in SAR^2 is set to 100 for vision tasks and 10 for LLM reasoning tasks. For trainable parameters of our methods during TTA, following Tent [8], we **adapt the affine parameters** of group/layer normalization layers for vision models and LoRA parameters for LLM. More details and hyperparameters of compared methods are put in Appendix A.2, available online.

A. Robustness to Corruption Under Wild Test Settings

Results under Online Imbalanced Label Distribution Shifts: As illustrated in Section VI, as the imbalance ratio q_{max}/q_{min} increases, TTA degrades more and more severe. Here, we make comparisons under the most difficult case: $q_{max}/q_{min}=\infty$, where test samples come in class order. We evaluate all methods under different corruptions via the same sample sequence for fairness.

From Table I, our SAR consistently outperforms Tent over the 15 corruption types on ResNet50-GN and VitBase-LN, suggesting its effectiveness. It is worth noting that Tent works well for many corruption types on VitBase-LN (*e.g.*, *defocus* and *motion blur*) and ResNet50-GN (*e.g.*, *pixel*), while consistently failing on ResNet50-BN. This further verifies our observations in Section VI that entropy minimization on LN/GN models has the potential to perform well under online imbalanced label distribution shifts. Meanwhile, Tent also suffers from many failure cases, *e.g.*, VitBase-LN on *shot noise* and *snow*. For these cases, our SAR works well. Moreover, our SAR^2 further substantially improves the robustness by incorporating the test-time feature regularization scheme, *e.g.*, increasing the average accuracy from 37.2% (SAR) to 51.7%, with a +7.8% gain compared to DeYO on ResNet50-GN in Table I. Notably, SAR^2 remains effective even in scenarios where all existing methods perform poorly (*e.g.*, +16.3% on *defocus* with R50-GN), suggesting the distinct strength of feature regularization.

TABLE III
COMPARISONS WITH STATE-OF-THE-ART METHODS ON IMAGENET-C (SEVERITY LEVEL 5) WITH **BATCH SIZE=1** REGARDING **ACCURACY (%)**. "BN"/"GN"/"LN" IS SHORT FOR BATCH/GROUP/LAYER NORMALIZATION. THE **BOLD** NUMBER INDICATES THE BEST RESULT.

	Noise			Blur				Weather				Digital				
Model+Method	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	Avg.
ResNet50 (BN)	2.2	2.9	1.9	17.9	9.8	14.8	22.5	16.9	23.3	24.4	58.9	5.4	17.0	20.6	31.6	18.0
• MEMO	7.5	8.7	8.9	19.7	13.0	20.8	27.6	25.4	28.7	32.2	60.9	11.0	23.8	32.9	37.5	23.9
• DDA	32.1	32.8	31.8	14.7	16.6	16.6	24.2	20.0	25.4	17.2	52.1	3.2	35.7	41.5	45.3	27.3
• Tent	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.2	0.1	0.1
• EATA	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.1	0.2	0.1	0.1	0.2	0.1	0.1
ResNet50 (GN)	18.0	19.8	17.9	19.8	11.4	21.4	24.9	40.4	47.3	33.6	69.3	36.3	18.6	28.4	52.3	30.6
• MEMO	18.5	20.5	18.4	17.1	12.6	21.8	26.9	40.4	47.0	34.4	69.5	36.5	19.2	32.1	53.3	31.2
• DDA	42.4	43.3	42.3	16.6	19.6	21.9	26.0	35.7	40.1	13.7	61.2	25.2	37.5	46.6	54.1	35.1
• Tent	2.5	2.9	2.5	13.5	3.6	18.6	17.6	15.3	23.0	1.4	70.4	42.2	6.2	49.2	53.8	21.5
• EATA	24.8	28.3	25.7	18.1	17.3	28.5	29.3	44.5	44.3	41.6	70.9	44.6	27.0	46.8	55.7	36.5
• DeYO	41.8	44.7	43.0	22.5	24.7	41.8	24.4	54.5	52.2	20.7	73.5	53.5	48.5	60.2	59.8	44.4
• ReCAP	42.5	44.4	42.9	19.4	25.0	42.2	44.0	49.7	52.4	57.5	72.9	53.6	29.5	60.4	60.0	46.4
• SAR (ours)	$23.4_{\pm 0.3}$	$26.6_{\pm 0.4}$	$23.9_{\pm 0.0}$	$18.4_{\pm 0.1}$	$15.4_{\pm 0.3}$	$28.6_{\pm 0.3}$	$30.4_{\pm 0.2}$	$44.9_{\pm 0.3}$	$44.7_{\pm 0.2}$	$25.7_{\pm 0.6}$	$72.3_{\pm 0.2}$	$44.5_{\pm 0.1}$	$14.8_{\pm 2.7}$	$47.0_{\pm 0.1}$	$56.1_{\pm 0.0}$	$34.5_{\pm 0.2}$
• SAR^2 (ours)	$\mathbf{46.2}_{\pm 0.2}$	$\mathbf{48.3}_{\pm 0.1}$	$\mathbf{46.7}_{\pm 0.1}$	$\mathbf{37.4}_{\pm 0.1}$	$\mathbf{38.6}_{\pm 0.3}$	$\mathbf{45.9}_{\pm 0.3}$	$\mathbf{50.9}_{\pm 0.1}$	$\mathbf{58.9}_{\pm 0.1}$	$\mathbf{56.6}_{\pm 0.1}$	$\mathbf{63.4}_{\pm 0.2}$	$\mathbf{73.9}_{\pm 0.0}$	$\mathbf{57.2}_{\pm 0.1}$	$\mathbf{57.5}_{\pm 0.1}$	$\mathbf{63.7}_{\pm 0.1}$	$\mathbf{62.4}_{\pm 0.1}$	$\mathbf{53.8}_{\pm 0.02}$
VitBase (LN)	9.5	6.7	8.2	29.0	23.4	33.9	27.1	15.9	26.5	47.2	54.7	44.1	30.5	44.5	47.8	29.9
• MEMO	21.6	17.3	20.6	37.1	29.6	40.4	34.4	24.9	34.7	55.1	64.8	54.9	37.4	55.4	57.6	39.1
• DDA	41.3	41.1	40.7	24.4	27.2	30.6	26.9	18.3	27.5	34.6	50.1	32.4	42.3	52.2	52.6	36.1
• Tent	42.2	1.0	43.3	52.4	48.2	55.5	50.5	16.5	16.9	66.4	74.9	64.7	51.6	67.0	64.3	47.7
• EATA	29.7	25.1	34.6	44.7	39.2	48.3	42.4	37.5	45.9	60.0	65.9	61.2	46.4	58.2	59.6	46.6
• DeYO	54.0	52.1	55.1	58.8	59.5	64.2	53.5	68.2	66.4	73.7	78.3	68.2	68.9	73.8	70.8	64.4
• ReCAP	53.5	56.7	56.9	59.2	60.5	65.3	64.0	69.6	67.2	74.1	78.4	64.6	70.2	74.4	71.5	65.7
• SAR (ours)	$40.8_{\pm 0.4}$	$36.4_{\pm 0.7}$	$41.5_{\pm 0.3}$	$53.7_{\pm 0.2}$	$50.7_{\pm 0.1}$	$57.5_{\pm 0.1}$	$52.8_{\pm 0.3}$	$59.1_{\pm 0.4}$	$50.7_{\pm 0.6}$	$68.1_{\pm 1.4}$	$74.6_{\pm 0.7}$	$65.7_{\pm 0.0}$	$57.9_{\pm 0.1}$	$68.9_{\pm 0.1}$	$65.9_{\pm 0.0}$	$56.3_{\pm 0.1}$
• SAR^2 (ours)	$\mathbf{57.2}_{\pm 0.0}$	$\mathbf{58.0}_{\pm 0.2}$	$\mathbf{58.1}_{\pm 0.2}$	$\mathbf{61.0}_{\pm 0.1}$	$\mathbf{62.0}_{\pm 0.1}$	$\mathbf{66.6}_{\pm 0.1}$	$\mathbf{65.8}_{\pm 0.1}$	$\mathbf{70.4}_{\pm 0.1}$	$\mathbf{68.8}_{\pm 0.1}$	$\mathbf{75.2}_{\pm 0.1}$	$\mathbf{79.1}_{\pm 0.1}$	$\mathbf{69.5}_{\pm 0.1}$	$\mathbf{71.4}_{\pm 0.2}$	$\mathbf{75.2}_{\pm 0.1}$	$\mathbf{72.0}_{\pm 0.1}$	$\mathbf{67.4}_{\pm 0.04}$

Results under Batch Size = 1: From Table III, our SAR mitigates the collapse issue in Tent and achieves robust performance gain in many cases, suggesting its effectiveness. It is worth noting that MEMO/DDA does not suffer from the instability issue and achieves the same results across different wild testing scenarios, by resetting model parameters when adapting to each sample. However, they require significantly more computational overhead (see Table II, rows 1-3) and yield only a limited performance gain as they cannot leverage knowledge from previously seen samples. In contrast, our SAR enables robust and efficient online adaptation through reliable and sharpness-aware updates. Furthermore, our SAR^2 unleashes the effectiveness and stability of TTA with the test-time feature regularization design, *e.g.*, increasing the average accuracy from 56.3% (SAR) to 67.4% and surpassing ReCAP by 1.7% on Vitbase-LN in Table III, consistently outperforming all baselines. Notably, SAR^2 also dramatically reduces performance variance (*i.e.*, by up to *10*-fold on ResNet50-GN compared to SAR) and remains effective even in scenarios where all existing methods underperform (*e.g.*, *defocus* and *glass* on R50-GN in Table III). These results highlight the crucial role of test-time feature regularization in online TTA, enabling SAR^2 to achieve both substantially higher accuracy and enhanced stability across diverse wild test settings.

Results under Mixed Distribution Shifts: We evaluate different methods on the mixture of 15 corruption types (total of 15 $\times$ 50,000 images) at different severity levels (5&3). From Table IV, our SAR achieves robust performance across models and severity levels, suggesting its effectiveness. Tent fails (occurs collapse) on ResNet50-GN levels 5&3 and VitBase-LN level 5 and achieves inferior accuracy than the no-adapt model, showing the instability of long-range online entropy minimization. Compared with Tent, although MEMO and DDA achieve better results, they rely on much more computation (inefficient at test time) as in Table II, and DDA also need to train an additional diffusion model. DeYO and ReCAP mitigate the collapse issue in Tent by selectively adapting to individual samples. However, they use solely output signals for sample selection, which remains insufficient and achieves a performance plateau on VitBase-LN (*e.g.*, 59.3%-59.4% at level 5). In SAR^2, by integrating test-time feature regularization, we break through the DeYO/ReCAP performance ceiling, improving over SAR by 5.0% on ResNet50-GN and 4.1% on VitBase-LN at level 5 in Table IV, consistently outperforming all baselines. These results suggest that regularizing the model representation at test time provides distinct robustness and effectiveness for TTA.

TABLE IV
COMPARISONS WITH STATE-OF-THE-ARTS ON IMAGENET-C UNDER **MIXTURE OF 15 CORRUPTION TYPES** REGARDING **ACCURACY (%)**

	ResNet50 (GN)		VitBase (LN)		
Method	Level 5	Level 3	Level 5	Level 3	Avg.
NoAdapt	30.6	54.0	29.9	53.8	42.1
MEMO [48]	31.2	54.5	39.1	62.1	46.7
DDA [76]	35.1	52.3	36.1	53.2	44.2
Tent [8]	13.4	33.1	16.5	70.2	33.3
EATA [12]	38.1	56.1	55.7	69.6	54.9
ROID [20]	36.9	56.9	52.9	68.2	53.7
DeYO [19]	38.6	59.2	59.4	72.1	57.3
ReCAP [54]	41.5	59.8	59.4	72.1	58.2
SAR (ours)	$38.3_{\pm 0.1}$	$57.4_{\pm 0.1}$	$57.1_{\pm 0.1}$	$70.7_{\pm 0.01}$	$55.9_{\pm 0.01}$
SAR^2 (ours)	$\mathbf{43.3}_{\pm 0.1}$	$\mathbf{59.8}_{\pm 0.01}$	$\mathbf{61.2}_{\pm 0.03}$	$\mathbf{72.9}_{\pm 0.01}$	$\mathbf{59.3}_{\pm 0.01}$

Results under Continuous Adaptation with Online Imbalanced Label Shifts: We further evaluate the long-term stability of TTA methods under a more challenging yet practical test scenario in the wild, where the model is continuously adapted across domains under online imbalanced label shifts, without parameter resets. From Table V, our SAR improves upon Tent by a large margin, suggesting our effectiveness, but it still suffers from instability during prolonged TTA. Prior methods such as DeYO and ReCAP also exhibit severe degradation over time, *e.g.*, collapsing with accuracy close to zero on *blurs* and *weathers* with ResNet50-GN. This indicates that while selecting

TABLE V
COMPARISONS WITH STATE-OF-THE-ART METHODS ON IMAGENET-C (SEVERITY LEVEL 5) UNDER **CONTINUOUS ADAPTATION** WITH ONLINE IMBALANCED LABEL SHIFTS (IMBALANCE RATIO $= \infty$) REGARDING **ACCURACY (%)**. "GN"/"LN" IS SHORT FOR GROUP/LAYER NORMALIZATION. THE **BOLD** NUMBER INDICATES THE BEST RESULT.

	Noise			Blur				Weather				Digital				
Model+Method	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	Avg.
ResNet50 (GN)	17.9	19.9	17.9	19.7	11.3	21.3	24.9	40.4	47.4	33.6	69.2	36.3	18.7	28.4	52.2	30.6
• MEMO	18.4	20.6	18.4	17.1	12.7	21.8	26.9	40.7	46.9	34.8	69.6	36.4	19.2	32.2	53.4	31.3
• DDA	42.5	43.4	42.3	16.5	19.4	21.9	26.1	35.8	40.2	13.7	61.3	25.2	37.3	46.9	54.3	35.1
• Tent	2.6	0.2	0.1	1.4	0.1	0.1	0.2	0.3	0.1	0.2	0.3	0.1	0.1	0.1	0.3	0.4
• EATA	25.8	23.7	20.2	14.4	13.7	18.4	20.2	20.8	23.2	32.0	41.4	20.5	22.4	23.1	26.5	23.1
• ROID	32.6	35.2	33.1	15.7	19.3	27.1	31.3	48.0	45.1	52.4	**72.0**	45.9	35.1	46.7	54.2	39.6
• DeYO	41.9	47.8	46.1	7.4	0.4	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	9.7
• ReCAP	42.2	47.8	46.6	5.8	0.2	0.1	0.1	0.2	0.2	0.1	4.3	**53.6**	15.4	44.8	58.3	21.3
• SAR (ours)	$33.7_{\pm 1.0}$	$46.1_{\pm 0.2}$	$46.6_{\pm 0.1}$	$10.9_{\pm 1.5}$	$11.9_{\pm 3.4}$	$14.1_{\pm 7.6}$	$3.0_{\pm 2.1}$	$25.6_{\pm 7.1}$	$17.8_{\pm 8.9}$	$38.7_{\pm 19.2}$	$70.6_{\pm 1.3}$	$48.6_{\pm 0.5}$	$5.7_{\pm 0.9}$	$0.9_{\pm 0.2}$	$36.7_{\pm 3.4}$	$27.4_{\pm 1.1}$
• SAR^2 (ours)	$\mathbf{43.2}_{\pm 0.5}$	$\mathbf{48.8}_{\pm 0.2}$	$\mathbf{47.8}_{\pm 0.2}$	$\mathbf{29.8}_{\pm 0.4}$	$\mathbf{35.2}_{\pm 0.2}$	$\mathbf{41.1}_{\pm 0.4}$	$\mathbf{46.2}_{\pm 0.4}$	$\mathbf{50.4}_{\pm 0.3}$	$\mathbf{53.1}_{\pm 0.2}$	$\mathbf{58.6}_{\pm 0.2}$	$69.1_{\pm 0.1}$	$50.9_{\pm 0.2}$	$\mathbf{48.8}_{\pm 0.4}$	$\mathbf{57.3}_{\pm 0.1}$	$\mathbf{59.1}_{\pm 0.1}$	$\mathbf{49.3}_{\pm 0.1}$
VitBase (LN)	9.4	6.7	8.3	29.1	23.4	34.0	27.0	15.8	26.3	47.4	54.7	43.9	30.5	44.5	47.6	29.9
• MEMO	21.6	17.4	20.6	37.1	29.6	40.6	34.4	25.0	34.8	55.2	65.0	54.9	37.4	55.5	57.7	39.1
• DDA	41.3	41.3	40.6	24.6	27.4	30.7	26.9	18.2	27.7	34.8	50.0	32.3	42.2	52.5	52.7	36.2
• Tent	32.5	0.3	0.1	0.4	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1	0.1	0.1	2.3
• EATA	36.1	27.7	22.4	14.7	20.0	13.4	5.7	8.2	11.4	6.6	24.2	0.3	10.2	15.6	19.2	15.7
• ROID	45.1	45.3	45.4	48.0	50.4	54.8	51.3	60.1	57.5	67.1	75.1	60.0	59.4	67.7	64.7	56.8
• DeYO	52.9	56.9	57.1	35.5	37.6	40.0	18.0	39.1	39.7	45.1	50.6	39.3	42.7	47.1	45.6	43.2
• ReCAP	53.3	56.7	57.2	53.8	57.1	60.9	39.1	61.1	62.8	69.3	76.3	62.3	**67.1**	71.4	68.7	61.1
• SAR (ours)	$47.2_{\pm 1.9}$	$54.0_{\pm 0.2}$	$56.2_{\pm 0.2}$	$53.0_{\pm 0.3}$	$55.8_{\pm 0.2}$	$59.0_{\pm 0.2}$	$56.2_{\pm 0.6}$	$61.1_{\pm 0.2}$	$63.1_{\pm 0.2}$	$68.8_{\pm 0.2}$	$76.4_{\pm 0.1}$	$63.7_{\pm 0.1}$	$60.8_{\pm 1.9}$	$69.7_{\pm 0.3}$	$67.8_{\pm 0.1}$	$60.9_{\pm 0.2}$
• SAR^2 (ours)	$\mathbf{53.7}_{\pm 0.1}$	$\mathbf{57.6}_{\pm 0.3}$	$\mathbf{58.2}_{\pm 0.1}$	$\mathbf{54.6}_{\pm 0.7}$	$\mathbf{58.4}_{\pm 0.2}$	$\mathbf{61.4}_{\pm 0.2}$	$\mathbf{60.7}_{\pm 0.4}$	$\mathbf{64.0}_{\pm 0.2}$	$\mathbf{64.3}_{\pm 0.2}$	$\mathbf{70.4}_{\pm 0.1}$	$\mathbf{76.7}_{\pm 0.1}$	$\mathbf{63.6}_{\pm 0.2}$	$66.6_{\pm 0.4}$	$\mathbf{71.8}_{\pm 0.3}$	$\mathbf{69.1}_{\pm 0.1}$	$\mathbf{63.4}_{\pm 0.1}$

pseudo labels for entropy minimization helps stabilize early TTA, model biases such as feature inequity may still accumulate and amplify during the long-term TTA process, hindering TTA stability. In contrast, by explicitly regulating feature redundancy and inequity, our SAR^2 prevents collapse and remains highly effective throughout continuous TTA, *e.g.*, increasing the average accuracy from 27.4% (SAR) to 49.3% with up to *10*-fold reduction in performance variance on ResNet50-GN in Table V. These further underscore the importance of test-time our feature regularization design in enabling robust TTA across both short-term and long-term adaptation in the wild.

Trigger Frequency of Model Recovery: We analyze when the model recovery/hard reset condition is triggered during online adaptation in this section. For SAR, resets are triggered only in a few challenging cases: for ViT-Base under *snow* and *frost* corruptions in online label shifts (Table I), for ViT-Base under *frost* corruption with batch size 1 (Table III), and for ResNet-50 under *snow*, *frost*, *fog*, or *JPEG* corruptions in online label shifts with continual adaptation (Table V), in which different runs may trigger at different corruptions. In contrast, for SAR^2, the reset condition is never triggered across all evaluated scenarios (Tables I, III, IV, and V). This indicates that SAR^2 substantially improves the intrinsic stability of TTA and does not rely on hard reset for its performance.

B. Ablate Experiments

Effects of Components in SAR: From Table VI, compared with plain entropy minimization, the reliable entropy in Eqn. (5) clearly improves the TTA performance by removing partial noisy updates, *e.g.*, $37.2\% \rightarrow 52.9\%$ w.r.t. the average accuracy over wild test settings. Meanwhile, sharpness-aware (sa) minimization in (6) promotes a flat entropy minima in the parameter space, which further enhances the robustness against large and noisy updates, *e.g.*, yielding a $+5.2\%$ improvement under TTA with a single sample. With both reliable entropy and sharpness optimization, our SAR performs stably except for very few cases, *e.g.*, on *snow* under label shifts (*see* Table F in Appendix, available online for more details). For this case, our model recovery scheme takes effect, *i.e.*, $55.8\% \rightarrow 57.1\%$ regarding the average accuracy under wild test scenarios.

TABLE VI
EFFECTS OF COMPONENTS IN SAR. WE REPORT **ACCURACY (%)** ON IMAGENET-C (LEVEL 5) WITH VITBASE-LN IN THE WILD. "RELIABLE (RE)" AND "SHARPNESS (SA)" DENOTE (5) & (6), "RECOVER (RC)" DENOTES MODEL RECOVERY SCHEME.

Exp.	RE	SA	RC	Label Shifts	Mix Shifts	$BS = 1$	Avg.
Tent				47.3	16.5	47.7	37.2
1	✓			53.1	55.2	50.5	52.9
2	✓	✓		54.5	57.1	55.7	55.8
3 (SAR)	✓	✓	✓	**58.0**	**57.1**	**56.3**	**57.1**

TABLE VII
EFFECTS OF COMPONENTS IN SAR^2. WE REPORT THE **ACCURACY (%)** ON IMAGENET-C (LEVEL 5) WITH VITBASE-LN UNDER WILD TEST SCENARIOS. FEATURE REDUNDANCY $R(\cdot)$ AND INEQUITY $I(\cdot)$ ARE CALCULATED BASED ON FEATURE BANK.

Exp.	$R(\cdot)$	$I(\cdot)$	Label Shifts	Mix Shifts	$BS = 1$	Avg.
SAR			58.0	57.1	56.3	57.1
1	✓		63.8	60.3	64.2	62.8
2		✓	63.3	58.9	61.1	61.1
3 (SAR^2)	✓	✓	**64.6**	**61.2**	**67.4**	**64.4**

Effects of Components in SAR^2: As discussed in Section V, inequity $I(\cdot)$ operates at the centroid level to prevent the representation bias towards a specific class. In contrast, redundancy $R(\cdot)$ is centroid-invariant, measuring inter-dimension feature correlations independent of the centroid. They thus focus complementarily on the centroid and centroid-invariant perspectives to maintain both balanced/unbiased centroids and decorrelated features during TTA. As shown in Table VII, each regularizer independently improves the efficacy of TTA in the wild, *e.g.*, 57.1% (SAR) *vs.* 62.8% (SAR + redundancy) *vs.* 61.1% (SAR

TABLE VIII
EFFECTS OF BATCH-IMBALANCE PENALTIES. RESULTS ARE REPORTED ON IMAGENET -C (LEVEL 5) UNDER A MILD SCENARIO W.R.T. **ACCURACY (%, ↑)** AND **ECE (%, ↓)**.

Method	ResNet50 (GN) Acc.	ResNet50 (GN) ECE	VitBase (LN) Acc.	VitBase (LN) ECE	Avg. Acc.	Avg. ECE
NoAdapt	30.6	8.4	29.9	7.0	30.3	7.7
Tent [8]	25.1	48.7	47.9	22.3	36.5	35.5
✢ InfoMax [38]	38.2	20.2	53.8	8.2	46.0	14.2
✢ Our $I(\cdot)$ (Eqn. 4)	48.9	7.5	59.3	4.9	**54.1**	**6.2**

TABLE IX
IMPACTS OF INITIAL REPRESENTATION HEALTH ON TTA EFFECTIVENESS. RESULTS ARE REPORTED ON IMAGENET-C (GAUSSIAN, LEVEL 5) UNDER LABEL SHIFTS REGARDING **ACCURACY (%)**. WE PERFORM K STEPS OF FEATURE REGULARIZATION (3)–(4) ON A RANDOM MINI-BATCH TO BOOST INITIAL REPRESENTATION HEALTH BEFORE TTA.

Model	Tent	SAR	DeYO	Avg.
ResNet50 (GN)	2.6	33.7	41.1	25.8
✢ feature regularizer ($K = 2$)	5.1	35.0	42.2	$27.4_{(+1.6)}$
✢ feature regularizer ($K = 4$)	16.3	36.9	42.8	$32.0_{(+6.2)}$
✢ feature regularizer ($K = 6$)	33.5	38.4	43.1	$\mathbf{38.4}_{(+12.6)}$

+ inequity) regarding average accuracy over wild scenarios. By combining both regularizers, the efficacy and stability of TTA are further improved by a large margin, *e.g.*, achieving an accuracy of 67.4% under TTA with a single sample in the 4th row. These results highlight both the effectiveness and the complementarity of our regularizer designs.

Comparison with Gradient Clipping: As mentioned in Section V-A, gradient clipping is a straightforward solution to alleviate model collapse. Here, we compare our SAR with two variants of gradient clip, *i.e.*, by value and by norm. From Fig. 8, for both two variants, it is hard to set a proper threshold δ for clipping, since the gradients for different models and test data have different scales and thus the δ selection would be sensitive. We carefully select δ on a specific test set (*shot noise* level 5). Then, we select a very small δ to make gradient clip work, *i.e.*, clip by value 0.001 and by norm 0.1. Nonetheless, the performance gain over "no adapt" is very marginal, since the small δ would limit the learning ability of the model and in this case the clipped gradients may point in a very different direction from the true gradients. However, a large δ fails to stabilize the adaptation process and the accuracy will degrade after the model collapses (*e.g.*, clip by value 0.005 and by norm 1.0). In contrast, SAR does not need to tune such a parameter and achieves significant improvements than gradient clipping.

Advantages of Inequity over Information Maximization [38]: We further compare our inequity regularizer $I(\cdot)$ with the diversity term in information maximization (IM) loss [81], which both penalize biased representation/prediction towards a specific class, while our $I(\cdot)$ operates at the representation's centroid level and IM loss operates at the output level. From Table VIII, although both inequity $I(\cdot)$ and IM loss alleviate the collapse issue in Tent, our inequity consistently outperforms IM loss regarding both accuracy and calibration across architectures, *e.g.*, increasing the accuracy from 46.0% to 54.1% and reducing ECE from 14.2% to 6.2%. This is because IM loss averages the softmax output of batch samples for entropy estimation, while $I(\cdot)$ first computes the centroid of batch features before calculating its prediction entropy. Under a limited batch size B, IM loss constrains the per-class total probability to B/C to achieve a uniform batch-average prediction, implying an implicit confidence flattening effect to TTA. In contrast, our inequity loss $I(\cdot)$ only ensures the representation's centroid is debiased, while each sample may retain a confident prediction. Detailed results across 15 domains (Table V in Appendix, available online) further highlight that IM loss can decrease Tent's accuracy on domains where Tent itself does not collapse, whereas our inequity $I(\cdot)$ consistently improves Tent's accuracy across all domains, suggesting our effectiveness for online TTA.

Impacts of Initial Representation Health on TTA: As shown in Fig. 3, a more severe distribution shift induces larger redundancy and inequity in model representation before TTA takes place. To further examine how this initial representation health impacts TTA efficacy, we conduct analysis in Table IX by applying feature regularization (3)–(4) to K random mini-batches before we start TTA. From the results, online TTA significantly benefits from better initial representation, *e.g.*, 25.8%→38.4% with feature regularizer ($K = 6$) w.r.t. average accuracy. Notably, poor initial health, *e.g.*, high inequity, can mislead online adaptation and worsen during TTA (as in Fig. 3), causing methods like Tent to collapse on the original ResNet50-GN. In contrast, by simply enhancing initial representation, Tent avoids collapse and achieves an accuracy of 33.5%. These highlight the crucial role of well-conditioned initial representations for stable TTA, and also the potential of our test-time feature regularization as a plug-in to existing methods.

Separability in Feature Space: We compare the separability in feature space based on t-SNE, where each point is color-coded according to its ground-truth label, and 20 classes are selected for visualization. Here, Tent uses a smaller learning rate to mitigate model collapse. From Fig. 7, we have the following observations: 1) Without adaptation, features are scattered and poorly separated; 2) Tent slightly improves separability but also produces a large and overly dense cluster. This coincides with the increase in feature inequity (Fig. 3), where representations from multiple classes gradually collapse toward a dominant one, leading to higher representation similarity; 3) SAR further improves separability compared to Tent, but several clusters remain highly similar; 4) SAR^2 incorporates the redundancy and inequity regularizers, which explicitly regulate representation overlap across classes and balance the spread of features, resulting in well-separated, distributed, and more compact clusters, thereby providing a more stable feature space for TTA.

Sharpness of Loss Surface: We visualize the loss surface by adding perturbations to model weights, as done in [82]. We plot Fig. 9 via the model weights obtained after the adaptation on the whole test set. By comparing Figs. 9(a) and (b), the area (the deepest blue) within the lowest loss contour line of our SAR is larger than Tent, suggesting that our solution achieves flatter entropy minima and thus is more robust to noisy/large gradients.



(a) VitBase-LN, NoAdapt (b) Tent [8] (c) SAR (ours) (d) SAR^2 (ours)

Fig. 7. t-SNE visualization of the feature space after adapting VitBase-LN on ImageNet-C (Snow, level 5) under online imbalanced label shifts.

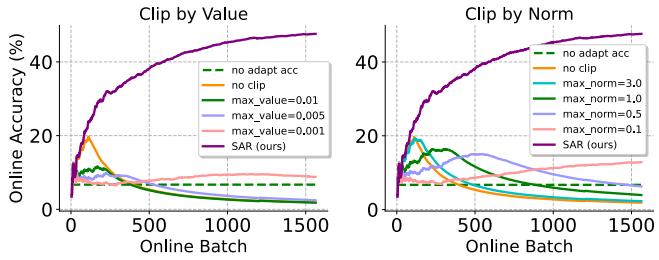


Fig. 8. Comparison with gradient clipping. Results on VitBase-LN, ImageNet-C, shot noise, severity level 5, online imbalanced (ratio $= \infty$) label shift. *Accuracy is calculated over all previous test samples.*

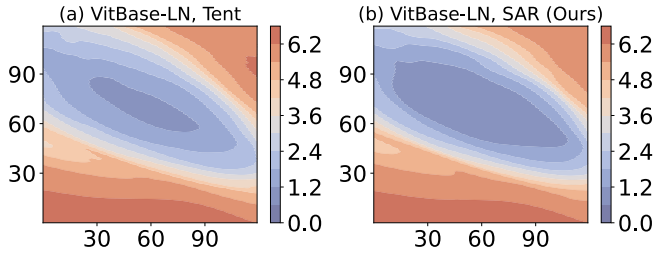


Fig. 9. Visualization of entropy-loss surface for VitBase-LN adapted via Tent and our SAR on ImageNet-C, Gaussian noise, severity level 5.

TABLE X
COMPARISONS WITH STATE-OF-THE-ART METHODS ON CORRUPTED IMAGENET-LT REGARDING **ACCURACY (%)**. WE ADD CORRUPTIONS TO IMAGENET-LT FOLLOWING IMAGENET-C WITH SEVERITY LEVEL 5. FOR SINGLE DOMAIN ADAPTATION, WE REPORT THE AVERAGE ACCURACY OVER 15 CORRUPTIONS. FOR CONTINUAL ADAPTATION, WE REPORT THE OVERALL ACCURACY OF 15 CORRUPTIONS.

	Single domain adaptation				Continual adaptation			
Model+Method	Many	Medium	Few	Overall	Many	Medium	Few	Overall
ResNet-GN	34.8	33.5	33.0	34.5	34.8	33.5	33.0	34.5
• Tent	25.0	24.6	24.4	24.6	0.6	0.4	0.3	0.4
• EATA	54.4	52.6	52.0	54.0	51.8	49.7	48.9	51.1
• ROID	53.8	51.9	52.1	53.2	54.1	52.1	52.1	53.5
• DeYO	52.0	50.3	49.5	51.7	12.0	11.5	11.0	12.0
• ReCAP	58.5	56.5	55.9	58.3	57.1	54.1	53.1	57.3
• SAR^2 (ours)	**61.8**	**59.6**	**58.5**	**61.5**	**60.0**	**57.4**	**56.5**	**60.0**
ViT-Base	33.1	31.5	31.5	32.9	33.1	31.5	31.5	32.9
• Tent	54.7	52.5	51.9	54.5	3.9	3.8	3.4	3.9
• EATA	69.3	66.4	65.3	69.1	68.5	65.1	63.4	68.2
• ROID	67.5	64.6	63.8	67.1	67.7	65.0	63.9	67.4
• DeYO	71.1	68.4	66.9	71.2	68.9	65.1	62.7	69.3
• ReCAP	71.9	69.1	67.7	71.9	70.1	66.4	64.1	70.4
• SAR^2 (ours)	**72.9**	**70.3**	**69.2**	**72.8**	**72.7**	**69.3**	**68.3**	**72.7**

TABLE XI
COMPARISONS ON IMAGENET-P WITH VIT-BASE REGARDING AVERAGE **ACCURACY** AND **MEAN FLIP RATE (MFR) OVER 10 SCENARIOS**

	No adapt	EATA	ROID	DeYO	ReCAP	SAR^2
Acc. (%, ↑)	73.6	40.1	26.0	78.0	79.2	**81.0**
mFR (%, ↓)	6.66	6.48	**1.25**	3.58	3.76	3.52

Effectiveness under Global Long-Tailed Distribution: Naturally long-tailed test streams are common in real-world deployments, where some classes may appear much more frequently than others. In this scenario, a naive uniform-prior regularizer that directly enforces class-balanced predictions can introduce harmful bias. In contrast, our inequity regularizer does not impose a class-balanced prior on sample-level predictions or mini-batch outputs. Instead, it first calculates feature centroids across *each* class, and applies regularization to class-level centroids that adaptively balances the contributions of majority and minority classes to the regularization, with the goal of preventing representations from collapsing toward a single dominant class during entropy-minimization-based TTA. To further verify this, we evaluate SAR^2 on ImageNet-LT under corrupted test streams. As in Table X, SAR^2 consistently improves TTA stability and achieves better performance across many-, medium-, and few-shot groups, suggesting that the regularizer prevents degenerate class collapse rather than enforcing an unrealistic balanced prediction prior. This mainly benefits from "Applying inequity regularization to partial class prototypes is sufficient to stabilize TTA against collapse" (as illustrated in Appendix D.1, available online).

Effectiveness under Dynamic Perturbations: We conduct experiments on ImageNet-P with ViT-Base, where the model is evaluated under sequential perturbations. As in Table XI, SAR^2 achieves the best average accuracy and competitive mFR. Notably, although ROID obtains a lower mFR, this is mainly because the model suffers from prediction collapse and tends to assign samples to incorrect but stable classes, as reflected by its much lower accuracy. In contrast, SAR^2 maintains both high accuracy and low fluctuation, suggesting stronger robustness under dynamic perturbations.

Effectiveness for Incentivizing Reasoning Capability Online: A statistical reasoning model may fail to generalize to the diverse reasoning tasks in real-world applications, while retraining the model offline is both expensive and time-consuming. In this

TABLE XII
COMPARISONS WITH STATE-OF-THE-ARTS FOR **ONLINE ADAPTIVE REASONING** IN MATHEMATICAL REASONING BENCHMARKS REGARDING **ACCURACY (%)**

Model+Method	Math-500	CollegeMath	AIME24	Minerva	Average
Llama3.1-8B	49.20	25.00	3.33	20.96	24.62
• Tent	50.00	24.17	6.67	20.59	25.36
• EATA	48.80	24.83	6.67	20.59	25.22
• COME	49.80	25.42	6.67	22.74	26.16
• TLM	50.00	25.58	6.67	19.49	25.44
• SAR^2 (ours)	51.80	25.75	13.33	23.16	28.51

section, we further explore adaptively incentivizing the reasoning capability in a large language model online. As shown in Table XII, SAR^2 consistently improves over the non-adapted baseline and existing TTA methods. In particular, it improves the average accuracy from 24.62% to 28.51%, achieving a 3.89% absolute gain over plain inference. These results show that our method is not limited to standard vision classification benchmarks, but can also provide meaningful gains in more challenging reasoning scenarios where test-time generalization is important.

VIII. CONCLUSION

In this paper, we seek to stabilize online test-time adaptation (TTA) under wild test settings, *i.e.*, mix shifts, small batch, and imbalanced label shifts. To this end, we first analyze and conduct extensive empirical studies to verify why wild TTA fails. Then, we point out that batch norm acts as a crucial obstacle to stable TTA. Meanwhile, though batch-agnostic norms (*i.e.*, group and layer norm) perform more stably under wild settings, they still suffer from many failure cases. We then investigate these failures by examining model gradients and feature behaviors before and after collapse during TTA. Our findings indicate that collapse occurrence is consistently characterized by a sharp gradient explosion followed by severe degradation, accompanied by substantial deterioration of feature representations. To address these, we first propose a sharpness-aware and reliable entropy minimization method (SAR) by suppressing the effect of certain noisy test samples with large gradients. Furthermore, we introduce two feature regularizers to boost SAR, supported by a prototype feature bank, to facilitate regularization in online wild test settings, namely SAR^2. Specifically, the redundancy regularizer reduces inter-dimensional correlations, while the inequity regularizer mitigates feature bias toward specific classes, thereby preventing representation deterioration. Extensive experimental results demonstrate the stability and efficiency of our SAR and SAR^2 under wild test settings.

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